

CODE-KO



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Final Approval

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Declaration

We hereby declare that this document “**Code-Ko**” neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers, especially our supervisor **Islam Abbasi**. If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

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Dedication

Our project is dedicated to our parents, seniors, friends, and our supervisor "Islam Abbasi" who has been our continual source of inspiration and whose support has helped this project succeed. This project would not have been possible without their love and support.

Acknowledgement

First of all, we are obliged to Allah Almighty the Merciful, the Beneficent and the source of all Knowledge, for granting us the courage and knowledge to complete this Project.

We are greatly indebted to our project supervisor “Islam Abbasi”. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

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Abstract

At this stage of technical life, the technology is increasing day by day. And Machines are replacing the work load of human. And this will making easy life for human.

As we know that, programming is becoming a very demanding field now a days. Number of programming students are increasing day by day. And their jobs are highly paid. So, we want to build a website in which our teachers can upload their content and student can learn from there content.

We also know that existing learning website in Pakistan are not enough capable to figure out Students queries in learning which they face during lecture which presents several challenges and limitations, sometime student use **hate speech** in comments and also students have very **less interaction** with teachers or teachers **don't even reply** mostly. So, there is no mechanism is provided in current systems. Therefore, in our website we have already deal with these types of **challenges**. So, in our website we provide such platform in which teacher and student can easily interact with each.

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Abstract

Code-Ko is a web-based platform designed to connect or interact with students and teachers easily with each other. We have built this website using **MERN Stack** technology. The platform is tailored only for students and teachers.

This platform allows teachers to upload their content like videos and help students to learn from their content. If the student has any queries he can ask the question to his /her teacher under the video (comment section). We have implemented **AI** sentiment analysis in the comment section which will remove only **abusive** comments.

On the other hand, if the student didn't get any reply from the teacher then he can also use our **chatbot** which will give answers according to their content.

Chapter 1: Introduction

Code-Ko is a web-based platform designed to connect or interact student and teacher easily with each other. The platform is tailored only for students and teachers. This platform allows teachers to upload their content and student can learn from there content. As the current E-learning websites or systems in Pakistan are insufficient in addressing the learning difficulties and inquiries of students during online lectures, which poses various challenges and constraints.

The **main problem** is being addressed here is that sometime student use **hate speech** in comments and also students have very **less interaction** with teachers or teachers **don't even reply** mostly. So, there is no mechanism is provided in current systems. Therefore, in our website we have already taken steps to tackle these **issues**.

On the other hand, Code-ko also provides convenience for students. It provides transparency, security, and efficiency, making it easier for student and teacher to communicate with each other. Ultimately, this would boost innovation and promote economic growth in Pakistan.

1.1 Goals and Objectives

Main and foremost objective of developing this system is that Students can learn their course, ask multiple questions without any hesitation.

Goals:

- To provide a web-based platform that enables easy interaction between students and teachers.
- To facilitate the sharing of educational content by teachers with their students.

Objectives:

- To address the lack of interaction between students and teachers that is dominant in current E-learning systems in Pakistan.
- To address the issue of hate speech in student comments.
- To provide a mechanism for addressing these issues that is not currently available in other E-learning platforms.

1.2 Scope of the Project

- Our website will be developed on MERN.
- We will develop a website that will provide platform for teachers to create content there and student will get learning facility.
- Teachers will upload their content without hesitations because we are not allowing bad comments on their content.
- Students will learn more affectionately as when he faces any difficulty he will reach out towards FAQs.
- It will increase the interaction between student and teachers.
- Problems will be solved with ML.

The platform would also be accessible and user-friendly, and simplified for both teacher and student.

Chapter 2: Literature Review

2.1 Introduction

Code-Ko is a web-based platform designed to connect or interact student and teacher easily with each other. The platform is tailored only for students and teachers. This platform allows teachers to upload their content and student can learn from there content. As the current E-learning websites or systems in Pakistan are insufficient in addressing the learning difficulties and inquiries of students during online lectures, which poses various challenges and constraints.

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On the other hand, Code-ko also provides convenience for students. It provides transparency, security, and efficiency, making it easier for student and teacher to communicate with each other. Ultimately, this would boost innovation and promote economic growth in Pakistan.

2.2 Background and Problem Elaboration

At this stage of technical life, the technology is increasing day by day. And Machines are replacing the work load of human. And this will making easy life for human.

As we know that, programming is becoming a very demanding field now a days. Number of programming students are increasing day by day. And their jobs are highly paid. So, we want to build a website in which our teachers can upload their content and student can learn from there content.

We also know that existing learning website in Pakistan are not enough capable to figure out Students queries in learning which they face during lecture which presents several challenges and limitations, sometime student use **hate speech** in comments and also students have very **less interaction** with teachers or teachers **don't even reply** mostly. So, there is no mechanism is provided in current systems. Therefore, in our website we have already deal with these types

of **challenges**. So, in our website we provide such platform in which teacher and student can easily interact with each.

2.3 Detailed Literature Review

2.3.1 Definitions

2.3.2 Related Research Work 1

2.3.3 Related Research Work 2

2.4 Literature Review Summary Table

The columns in the table depend upon your problem and should be specific to your project.

Table 1: History of Computing Devices
The summary of various computing devices invented in the past from 1833-1901 is presented here.

No.	Name, reference	Inventor	Year	Input	Output	Description
1.	Analytical Engine, [1]	Charles Babbage	1833	Punch cards	Printer, curve plotter, bell	First general-purpose computer that had an arithmetical logic unit and could compute using conditional branching and loops. Also incorporated integrated memory.

2.5 Research Gap

2.6 Problem Statement

Chapter 3: Requirements and Design

In this chapter we have developed our functional requirements for our actors i.e. (**Student**, **Admin** and **Teacher**). The requirements are designed for specially for Code-Ko platform.

Code-Ko is basically a web-based platform designed to connect or interact with students and teachers easily with each other.

The platform is user-friendly, easy to navigate and search, and provide a convenient and efficient way for both parties to connect and interact with each other.

We created our system **use cases** against each functional requirement and also created use case diagrams, fully dressed use cases for our actors i.e. (User, Admin and Teacher).

3.1 Requirements:

3.1.1 Functional Requirements

➤ User / Student:

ID	Requirements
FR-1.1	User shall be able to Sign Up.
FR-1.2	User shall be able to login to their account.
FR-1.3	User shall be able to create their profile.
FR-1.4	User shall be able to edit their profile.
FR-1.5	User shall be able to recover password.
FR-1.6	Student shall be able to enroll their course itself.
FR-1.7	User shall be able to search for specific content.
FR-1.8	Students shall be able to post comments and ask questions under each content
FR-1.9	Students shall be able to rate and provide feedback on the content.
FR-1.10	Students shall be able to ask questions to the chatbot related to the content. The chatbot will provide relevant answers.
FR-1.11	User shall be able to receive notifications such as, responses to their comments.

➤ Teacher:

ID	Requirements
FR-2.1	Teacher shall be able to Register an account.
FR-2.2	Teacher shall be able to login to their account.
FR-2.3	Teacher shall be able to create their profile.
FR-2.4	Teacher shall be able to edit their profile.
FR-2.5	Teacher shall be able to view their profile.
FR-2.6	Teacher shall be able to respond queries raised by students through comments.
FR-2.7	Teacher shall be able to upload content.
FR-2.8	Teacher shall be able to manage their uploaded content like edit, delete or organized.

➤ Admin:

ID	Requirements
FR-3.1	Admin shall be able to login to the system.
FR-3.2	Admin shall be able to Add students, teachers.
FR-3.3	Admin shall be able to view user's profiles.
FR-3.4	Admin shall be able to remove user from the system.
FR-3.5	Admin also manage user roles and permissions.
FR-3.6	Admin shall be able manage user accounts like Create, Update, Delete user accounts.
FR-3.7	Admin shall be able to manage content upload by teachers.
FR-3.8	Admin shall be able to review and approve or reject content uploaded by teachers.
FR-3.9	Admin shall be able to monitor comments posted by students.
FR-3.10	Admin shall be able to configure and manage the chatbot functionality.
FR-3.11	Admin shall be able to review bad reports.

3.1.2 Non-Functional Requirements

- **User-Friendly Interface:** The User-Interface of the system is very simple and user-friendly, allowing users to navigate easily and access the desired features without any confusion.

3.1.3 Hardware and Software Requirements

Hardware Requirements

- **Server:** A dedicated server or cloud hosting service to host the website and handle user requests and data storage.
- **Storage:** Sufficient storage space to store uploaded content, user profiles, comments, and other data.
- **Processing Power:** Adequate processing power to handle concurrent user requests, content uploads, and AI comment filtering.

Software Requirements:

- **Operating System:** The server should run a compatible operating system such as Linux, Windows Server, or macOS Server.
- **Web Server:** A web server software like Apache or Nginx to serve web pages and handle HTTP requests.
- **Database:** In database we have used MongoDB to store user information, content, comments, and other data.
- **Programming Languages:** The website can be built using the MERN stack, which includes:
 - **JavaScript:** for server-side and client-side scripting.
 - **Node.js:** a JavaScript runtime environment for server-side development.
 - **Express.js:** a web application framework for building the server-side application.
 - **React.js:** a JavaScript library for building the client-side user interface.
- **AI Comment Filtering:** An AI framework or library to implement the comment filtering system, such as TensorFlow or PyTorch.
- **Chatbot:** A natural language processing (NLP) library or framework to implement the chatbot feature, such as Dialogflow or Rasa.

- **Additional Libraries and Frameworks:** Depending on specific requirements, additional libraries and frameworks may be needed to implement features such as user authentication, file uploading, notifications, and real-time communication.
- **Development Tools:** Integrated Development Environments (IDEs) like Visual Studio Code, code editors, and debugging tools for development and testing.
- **Version Control:** A version control system like Git to manage source code and collaborate with multiple developers.
- **Deployment:** Tools and services for deploying the website to the server or cloud hosting platform, such as Docker or AWS Elastic Beanstalk.

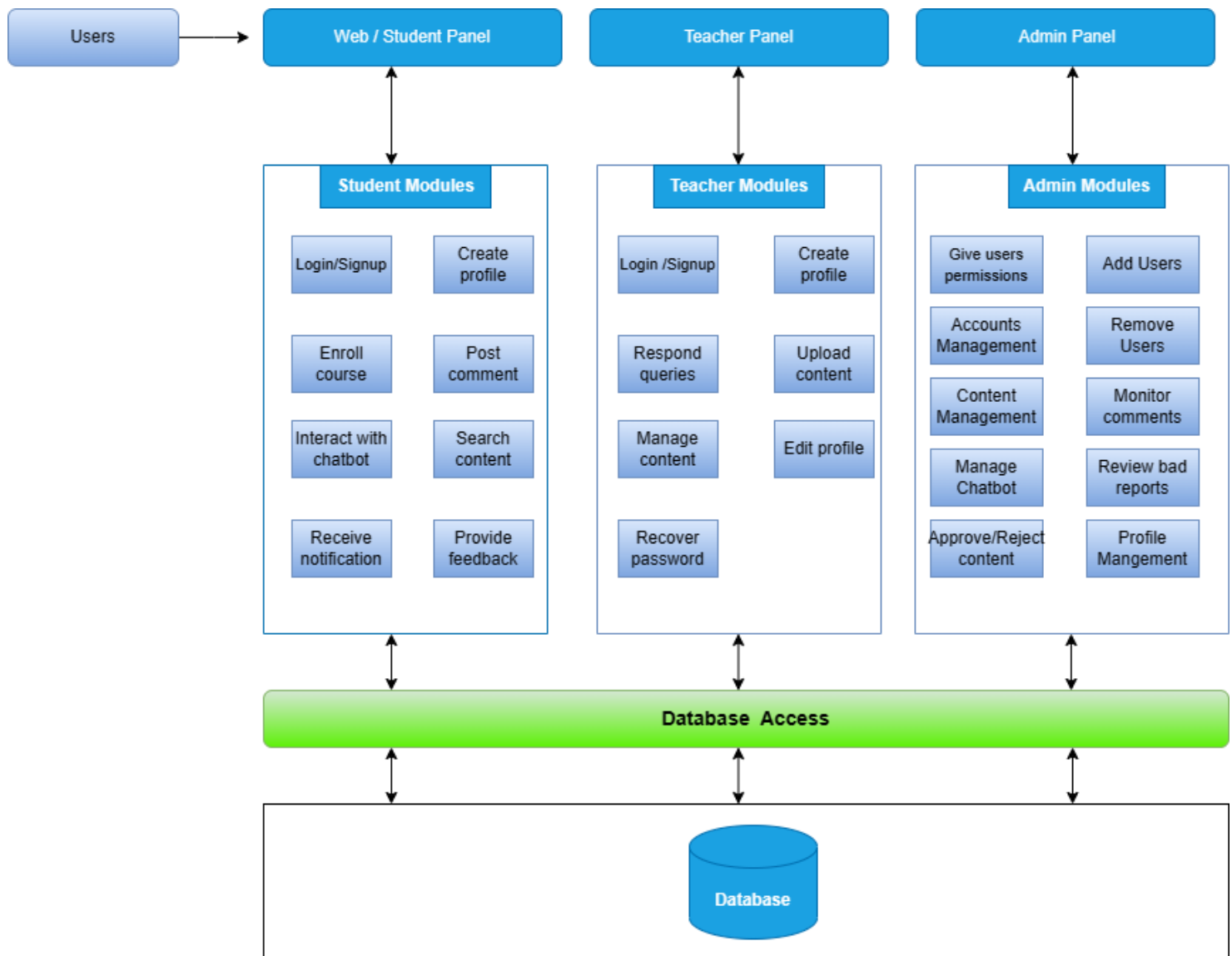
3.2 Proposed Methodology

Code-Ko is a web-based platform designed to connect or interact student and teacher easily with each other. The platform is tailored only for students and teachers. This platform allows teachers to upload their content and student can learn from there content. As the current E-learning websites or systems in Pakistan are insufficient in addressing the learning difficulties and inquiries of students during online lectures, which poses various challenges and constraints.

We are building this using MERN Stack technology. The platform is tailored only for students and teachers. This platform allows teachers to upload their content like videos and help students to learn from their content. If the student has any queries he can ask the question to his /her teacher under the video (comment section). We have implemented AI sentiment analysis in the comment section which will remove only abusive comments.

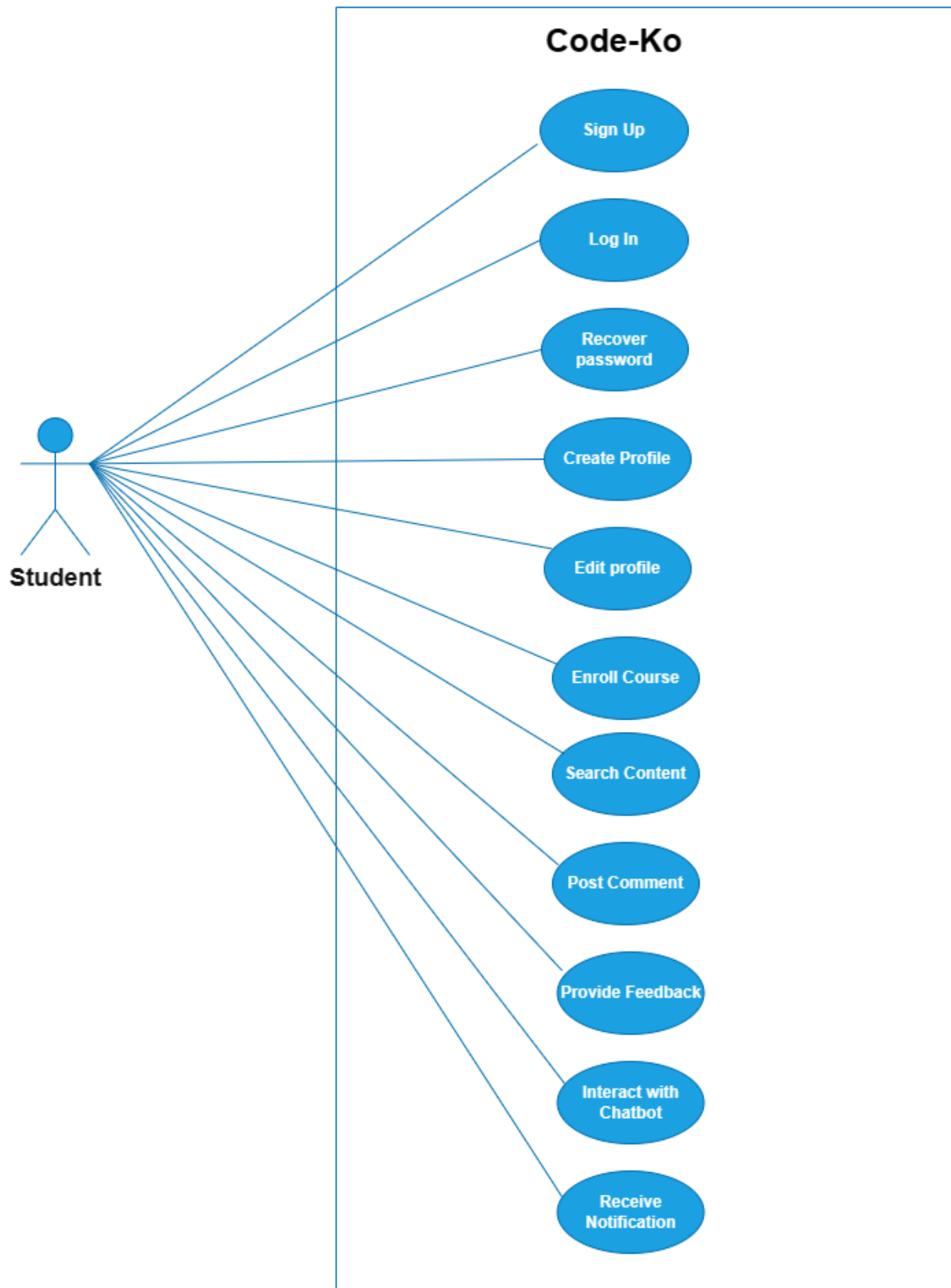
On the other hand, if the student didn't get any reply from the teacher then he can also use our chatbot which will give answers according to their content.

3.3 System Architecture:

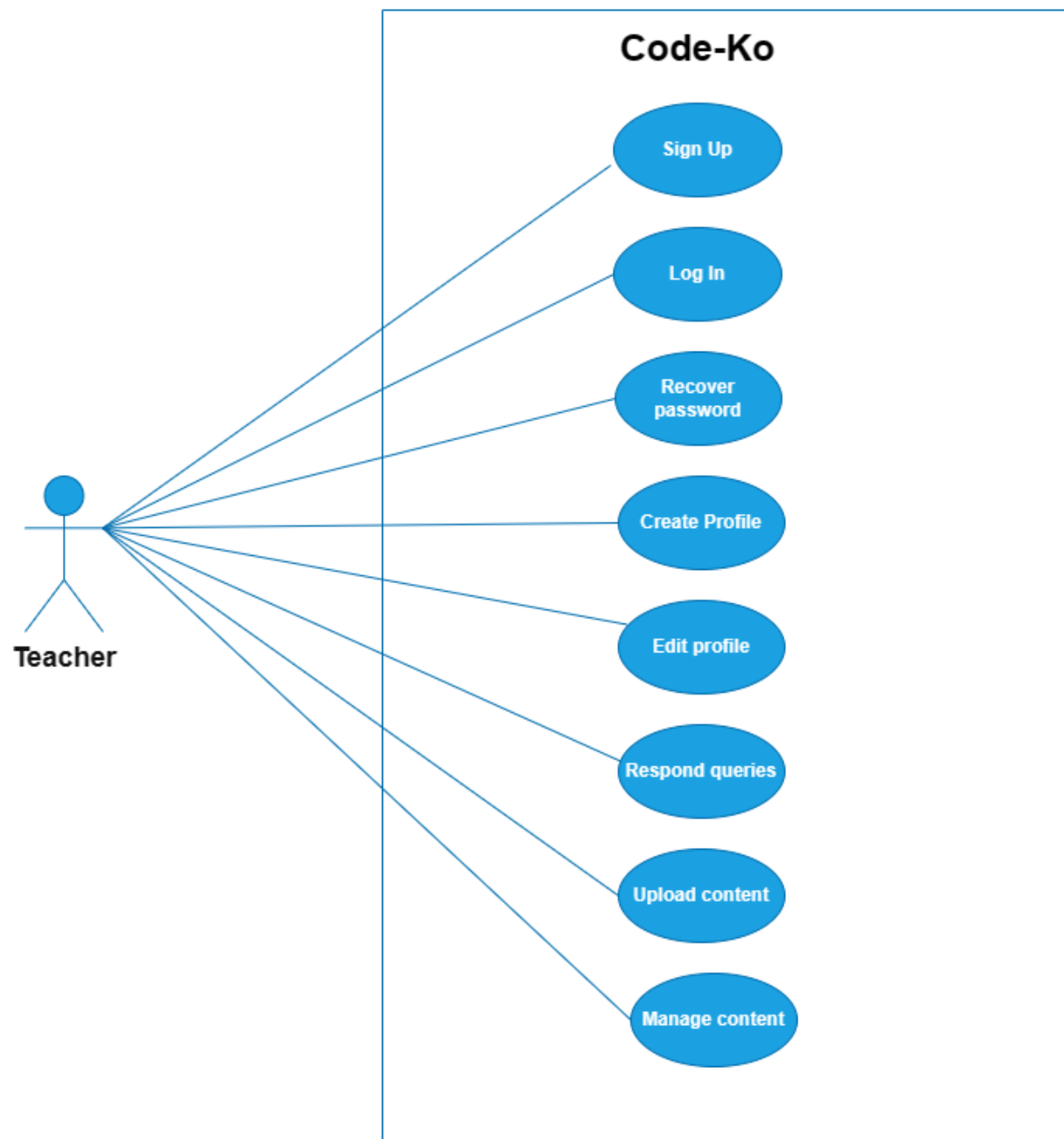


3.4 Use Cases:

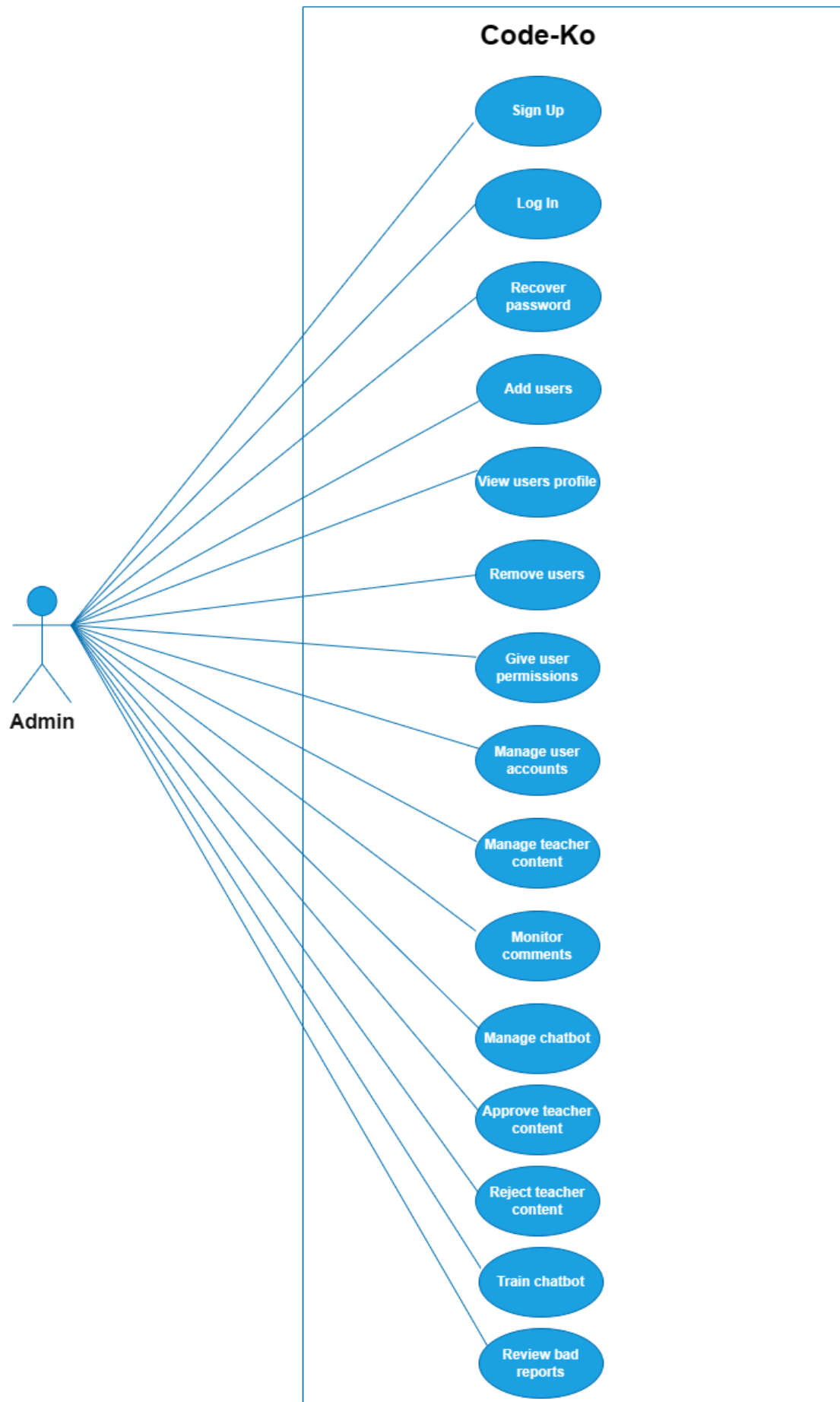
- **Student:**

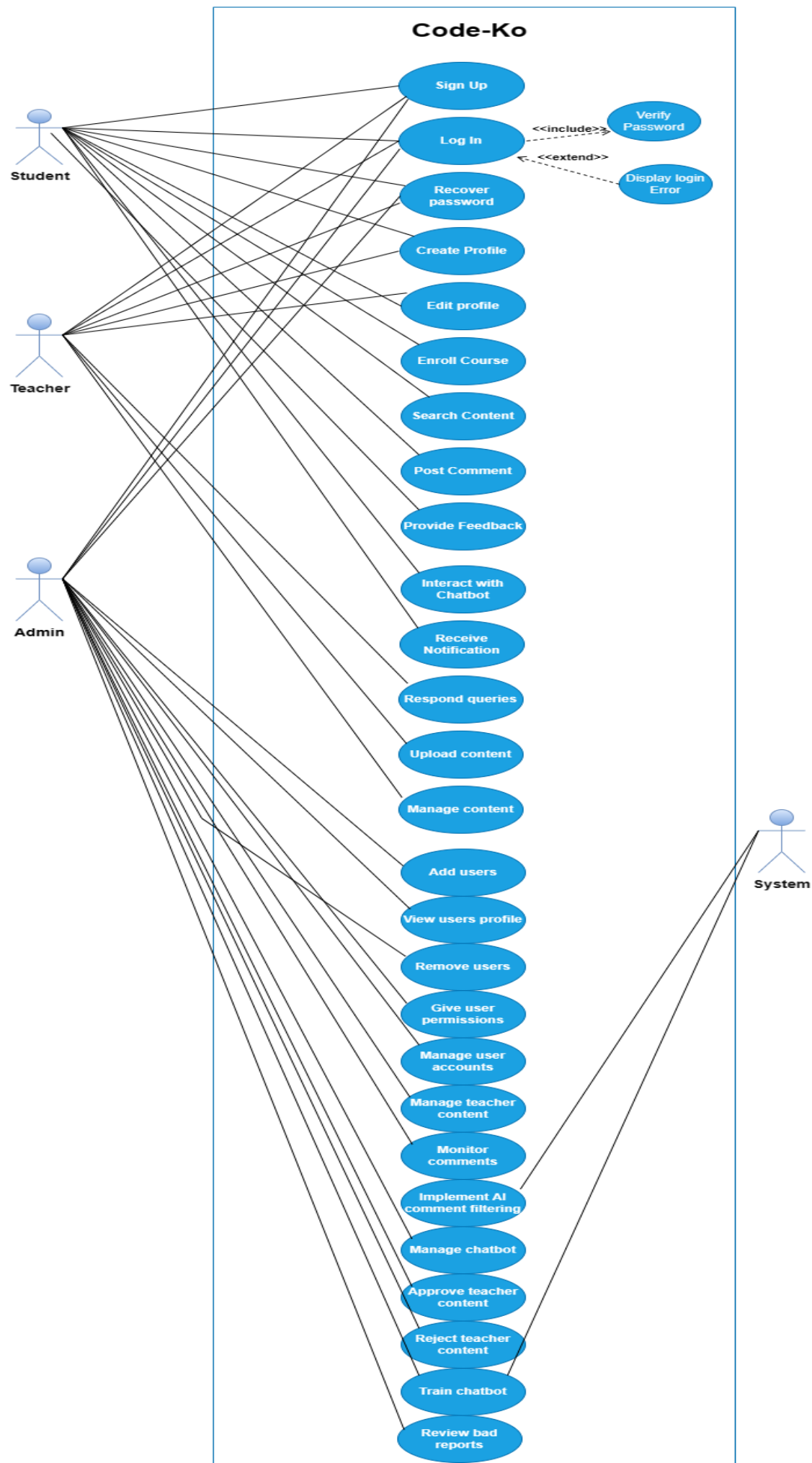


- **Teacher:**



- **Admin:**





Fully-Dressed Use Cases:

3.4.1 Login:

Name	Login		
Actors	Admin, Teacher, Student		
Summary	The user shall provide their email and password on the login form and after successful verification, redirect the user to the home page.		
Pre-Conditions	User must be registered on the system.		
Post-Conditions	The user shall be redirected to the home page of the system and user’s session will be maintained.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
3	The user enters valid email and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: <i>Incorrect email or password entered.</i>

3.4.2 Sign Up:

Name	Sign up
Actors	Teacher, Student, Admin
Summary	The user creates a new account by providing necessary information.
Pre-Conditions	None
Post-Conditions	The user successfully creates a new account and can log in to the system.
Special Requirements	None
Basic Flow	

Actor Action		System Response	
1	The user opens the sign-up page.	2	The sign-up page is displayed, asking for the user's information such as name, email, and password.
3	The user enters their name, email, and password.		
4	The user submits the sign-up form.	5	The system verifies the provided information and creates a new account for the user.
		6	The system displays a success message, informing the user that their account has been created.
Alternative Flow			
4.1	If the user submits the sign-up form with incomplete or invalid information.	4.2	The system responds with an <i>error message</i> , indicating the specific fields that need to be corrected.

3.4.3 Recover password:

Name	Recover password		
Actors	Student, Teacher		
Summary	This use case describes the process of recovering a password for a user who has forgotten or lost their login credentials.		
Pre-Conditions	The user must have a registered account on the Code-Ko platform. The user must have forgotten or lost their password.		
Post-Conditions	The user's password is successfully reset, and they can log in to their account using the new password.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko login page.		
2	User clicks on the "Forgot Password" link.	3	The system presents a password recovery form.
4	User enters their registered email address in the provided field.		
5	User clicks on the "Submit" button.	6	The system verifies the email address and generates a unique password reset token.
		7	The system sends an email to the user's registered email address containing a

			password reset link.
8	User checks their email inbox and clicks on the password reset link.	9	The system validates the password reset token and redirects the user to a password reset form.
10	User enters a new password and confirms it in the provided fields.		
11	User submits the password reset form.	12	The system verifies the new password and updates it in the user's account.
		13	The system displays a confirmation message indicating that the password has been successfully reset.
Alternative Flow			
4.1	If the user enters an invalid or unregistered email address.	4.2	The system displays an error message and prompts the user to enter a valid email address.
		4.3	If the password reset token has expired or become invalid, the system prompts the user to initiate the password recovery process again.

3.4.4 Create profile:

Name	Create Profile		
Actors	Student, Teacher		
Summary	This use case describes the process of creating a profile for a user on the Code-Ko platform.		
Pre-Conditions	The user must have access to the Code-Ko website. The user must be registered on the Code-Ko platform.		
Post-Conditions	The user's profile is successfully created and can be viewed by others on the platform.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko website.		
2	User clicks on the "Create Profile" or "Sign Up" button.	3	The system presents a profile creation form.
4	User enters their personal details, such as name, email address, username, educational background, expertise, bio, profile picture etc., in the provided fields.		

5	User clicks on the "Submit" or "Create Profile" button.	6	The system validates the entered information.
		7	The system saves the user's profile information in the Code-Ko database.
		8	The system displays a confirmation message indicating that the profile has been successfully created.
Alternative Flow			
4.1	If the user enters incomplete or invalid information in the profile creation form.	4.2	Then the system displays error messages and prompts the user to correct the entries.
4.3	If the chosen username is already taken by another user.	4.4	Then the system prompts the user to choose a different username.

3.4.5 Enroll Course:

Name		Enroll Course	
Actors		Student	
Summary		This use case describes the process of enrolling in a course on the Code-Ko platform.	
Pre-Conditions		The user must have a registered account on the Code-Ko platform. The user must have access to the desired course on the Code-Ko platform.	
Post-Conditions		The user is successfully enrolled in the selected course and gains access to its content and resources.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko website.		
2	User searches or browses the course catalog to find the desired course.		
3	User selects the desired course from the search results or catalog listing.	4	The system displays the course details, including the course description, instructor information.
5	User reviews the course details and decides to enroll.		
6	User clicks on the "Enroll" or "Join Course" button.	7	The system adds the user to the course.
		8	The system displays a confirmation message indicating that the user has been successfully enrolled in the course.
9	User can now access the course videos,		

	participate in any course-related activities.		
Alternative Flow			
6.1	If there are any technical issues during the enrollment process, such as server errors or connectivity problems.	6.2	The system displays an error message and advises the user to try again later.

3.4.6 Search Content:

Name		Search Content	
Actors		Student, Teacher	
Summary		This use case describes the process of searching for content on the Code-Ko platform.	
Pre-Conditions		- The user must have access to the Code-Ko website. - The user must be logged in to their Code-Ko account.	
Post-Conditions		The user receives search results based on their query and can access the relevant content.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko website and logs in to their account.		
2	User locates the search bar or search feature on the platform.		
3	User enters a keyword or query related to the content they are searching for.		
4	User clicks on the "Search" button or presses the "Enter" key.	5	The system processes the search query and retrieves relevant content matching the keyword or query.
		6	The system presents the search results to the user, displaying relevant content titles, descriptions.
7	User scans through the search results to find the desired content.	8	The system navigates the user to the selected content, displaying its full details or opening it for viewing.
9	User can now access and interact with the searched content, such as watching		

	videos.		
Alternative Flow			
3.1	If the user doesn't get the desired result.	3.2	Then the system may suggest user alternative search terms or provide recommendations for similar content.

3.4.7 Post Comment:

Name		Post Comment	
Actors		User (Student or Teacher)	
Summary		This use case describes the process of posting a comment on the Code-Ko platform.	
Pre-Conditions		• The user must have a registered account on the Code-Ko platform. • The user must be logged in to their Code-Ko account. • The user must be viewing content (e.g. video) where commenting is allowed.	
Post-Conditions		The user's comment is successfully posted and becomes visible to others who access the same content.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko website and logs in to their account.		
2	User accesses the content (e.g., video where they want to post a comment.		
3	User scrolls to the comment section or finds the designated area for posting comments.		
4	User clicks on the "Post Comment" or "Add Comment" button.	5	The system presents a comment form or text box.
6	User enters their comment in the provided text area.		
7	User clicks on the "Submit" or "Post" button to publish the comment.	8	The system adds the comment to the comment section of the respective content.
		9	The system displays a confirmation message indicating that the comment has been successfully posted.
10	User's comment becomes visible to others who access the same content.		

Alternative Flow			
6.1	If user enters inappropriate or abusive comment.	6.2	Then the system may have implemented AI sentiment analysis to identify and remove abusive or inappropriate comments.

3.4.8 Provide Feedback:

Name	Provide feedback		
Actors	User (Student or Teacher)		
Summary	This use case describes the process of providing feedback on the Code-Ko platform.		
Pre-Conditions	• The user must have a registered account on the Code-Ko platform. • The user must be logged in to their Code-Ko account. • The user must have accessed course, or content on the platform.		
Post-Conditions	The user's feedback is successfully submitted and received by the Code-Ko platform.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko website and logs in to their account.		
2	User accesses the course, or content for which they want to provide feedback.		
3	User locates the feedback section or finds the designated area for providing feedback.		
4	User clicks on the "Give Feedback" button.	5	The system presents a feedback form or text box.
6	User enters their feedback in the provided text area, providing specific details and information.		
7	User clicks on the "Submit" or "Send" button to submit the feedback.	8	The system captures and stores the user's feedback in the Code-Ko platform.
		9	The system displays a confirmation message indicating that the feedback has been successfully submitted.
Alternative Flow			

3.4.9 Interact with Chatbot:

Name		Interact with Chatbot	
Actors		Student	
Summary		This use case describes the process of interacting with the Chatbot on the Code-Ko platform.	
Pre-Conditions		• The user must have access to the Code-Ko website. • The Chatbot feature must be implemented and available on the platform.	
Post-Conditions		The user receives responses and assistance from the Chatbot based on their queries.	
Special Requirements		• The platform should have a Chatbot integrated using Natural Language Processing (NLP) techniques. • The Chatbot should be capable of understanding and responding to user queries or requests.	
Basic Flow			
Actor Action		System Response	
1	User navigates to the Code-Ko website.		
2	User locates the Chatbot feature, such as a chat window or a dedicated Chatbot section.		
3	User initiates the interaction with the Chatbot by typing a message or a query in the chat input field.		
4	User submits the message by pressing the "Enter" key or clicking the "Send" button.	5	The Chatbot system receives the user's message and processes it using NLP algorithms.
		6	The Chatbot analyzes the user's query and generates a response based on its programmed logic or knowledge base.
		7	The Chatbot displays the response in the chat window or replies to the user's message.
8	User reviews the Chatbot's response and may continue the conversation by asking follow-up questions or providing additional information.	9	The Chatbot continues to analyze and respond to the user's messages, providing relevant information.
10	The user and Chatbot engage in a conversation until the user's queries or requests are adequately addressed.		
Alternative Flow			
8.1	If the user doesn't get any desired response or Chatbot unable to	8.2	Then it may prompt the user to rephrase the question or seek clarification.

	understand the user's query.		
			If there are any technical issues with the Chatbot, such as system errors or connectivity problems, the platform may display an error message and advise the user to try again later.

3.4.10 Respond queries:

Name		Respond queries	
Actors		Teacher	
Summary		This use case describes the process of a teacher responding to queries raised by students through comments on the Code-Ko platform.	
Pre-Conditions		• The teacher must have a registered account on the Code-Ko platform. • The teacher must be logged in to their Code-Ko account. • The teacher must have access to the content where students have posted queries.	
Post-Conditions		The teacher successfully responds to the queries raised by students.	
Special Requirements		The platform should have a comment section or a designated area for students to post queries. The teacher's response should be visible to the students who posted the queries.	
Basic Flow			
Actor Action		System Response	
1	Teacher logs in to their Code-Ko account.		
2	Teacher navigates to the content or video where students have posted queries.		
3	Teacher reads through the comments section to identify queries posted by students.		
4	Teacher selects a specific query to respond to.		
5	Teacher types the response in the comment section, addressing the student who posted the query.		
6	Teacher clicks on the "Submit" or "Post" button to post the response.		
		7	The system validates the response and adds it to the comment section, associating it with the corresponding query.
		8	The system displays a confirmation

			message indicating that the response has been successfully posted.
		9	The student who posted the query receives a notification informing them of the teacher's response.
		10	The student can now view the teacher's response and engage in further discussion if needed.
Alternative Flow			
	The teacher may have the option to edit or delete their response after posting it.		Then the system can provide appropriate actions for modifying or removing responses.

3.4.11 Receive Notification:

Name		Receive Notification	
Actors		Teacher, Student	
Summary		This use case describes the process of receiving notifications on the Code-Ko platform, specifically for receiving responses to the user's comments.	
Pre-Conditions		• The user must be logged in to their Code-Ko account. • The user must have posted a comment on a piece of content.	
Post-Conditions		The user receives notifications when there are responses to their comments.	
Special Requirements		The platform should have a notification system implemented.	
Basic Flow			
Actor Action		System Response	
1	User logs in to their Code-Ko account.		
2	User navigates to the content or video where they have previously posted a comment.		
3	User views the content and scrolls to the comment section.		
4	Student or the instructor posts a response to the user's comment.	5	The system identifies the response and recognizes that it is directed towards the user.
		6	The system generates a notification for the user, indicating that there is a new response to their comment.
7	The user receives the notification through their preferred notification channel e.g., email, in-platform notification.		
8	User clicks on the notification to view the response.	9	The system redirects the user to the relevant content or comment section,

			highlighting the response to their comment.
10	User can now view and interact with the response, such as replying or engaging in a discussion.		
Alternative Flow			
	If there are any technical issues with the notification system, such as delays or failures in delivering notifications.		Then the system may display an error message and advise the user to check their notifications later.

3.4.12 Upload Content:

Name	Upload content		
Actors	Teacher		
Summary	This use case describes the process of a teacher uploading content to the Code-Ko platform.		
Pre-Conditions	• The teacher must have a registered account on the Code-Ko platform. • The teacher must be logged in to their Code-Ko account. • The teacher must have the necessary permissions to upload content.		
Post-Conditions	The teacher's content is successfully uploaded and made available on the Code-Ko platform.		
Special Requirements	The platform should have a designated area for teachers to upload content.		
Basic Flow			
Actor Action		System Response	
1	Teacher logs in to their Code-Ko account.		
2	Teacher navigates to the content management or upload section of the Code-Ko platform.		
3	Teacher selects the option to upload content.	4	The system presents a file selection dialog or an interface to choose the content files.
5	Teacher selects the content files from their device, such as videos other relevant materials.	6	The system validates the selected files.
7	Teacher provides necessary information about the content, such as title, description, or tags.		
8	Teacher clicks on the "Upload" or "Submit" button to initiate the upload process.	9	The system starts uploading the content files to the Code-Ko platform's storage.
		10	Once the upload is complete, the system notifies the teacher of the successful upload.
		11	The uploaded content is now accessible

			and available to students on the Code-Ko platform.
Alternative Flow			
8.1	If teacher faces any issues during upload process such as network failures or file error.	8.2	Then the system displays an error message and advises the teacher to retry the upload.
	The teacher may have the option to edit or update the content information after uploading it.		

3.4.21 Add Users:

Name		Add Users	
Actors		Admin, Teacher	
Summary		This use case describes the process of adding users to the Code-Ko platform, specifically by the administrator or teacher.	
Pre-Conditions		• The actor (Administrator or Teacher) must have the necessary privileges and permissions to add users. • The actor must be logged in to their Code-Ko account.	
Post-Conditions		The new users are successfully added to the Code-Ko platform.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	Actor logs in to their Code-Ko account with appropriate privileges (Administrator or Teacher).		
2	Actor navigates to the user management or account administration section of the Code-Ko platform.		
3	Actor selects the option to add users or create new accounts.	4	The system presents a form or interface to enter user details, such as email address, username, and other relevant information.
5	Actor fills in the required user information, ensuring the accuracy and completeness of the data.	6	The system validates the entered information, checking for any potential errors or conflicts (e.g., duplicate email addresses, invalid usernames).
7	Actor submits the form or clicks on the "Add User" button to initiate the user creation process.	8	The system processes the user creation request and generates a new user account.
		9	The system displays a confirmation message indicating that the user has been

			successfully added to the Code-Ko platform.
10	The new user receives an email or notification containing their login credentials and instructions for accessing the platform.		
Alternative Flow			
7.1	If there are any issues during the user creation process, such as validation errors or database constraints.	7.2	the system displays an error message and advises the actor to review and correct the provided information

3.4.22 Remove Users:

Name	Remove user		
Actors	Admin		
Summary	This use case describes the process of removing users from the Code-Ko platform. The removal can only be performed by the administrator.		
Pre-Conditions	The administrator must have the necessary privileges and permissions to remove users. The administrator must be logged in to their Code-Ko account.		
Post-Conditions	The selected users are successfully removed from the Code-Ko platform.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	Administrator logs in to their Code-Ko account with appropriate privileges.		
2	Administrator navigates to the user management or account administration section of the Code-Ko platform.		
3	Administrator views the list of users on the platform.		
4	Administrator selects one or multiple users to be removed.		
5	Administrator confirms the selection and initiates the removal process.	6	The system prompts the administrator to confirm the removal action, ensuring they understand the consequences.
7	Administrator confirms the removal action.	8	The system processes the removal request and deletes the selected users' accounts from the Code-Ko platform.

		9	The system displays a confirmation message indicating that the users have been successfully removed.
Alternative Flow			
	If there are any issues during the user removal process, such as database errors or system constraints.		The system displays an error message and advises the administrator to retry.

3.4.23 User permission:

Name	User permission		
Actors	Admin		
Summary	This use case describes the process of granting permissions to users on the Code-Ko platform. The permissions allow users to access specific features or perform certain actions based on their assigned roles.		
Pre-Conditions	The administrator must have the necessary privileges and permissions to grant user permissions. The administrator must be logged in to their Code-Ko account.		
Post-Conditions	The selected users have the assigned permissions and can perform the designated actions on the Code-Ko platform.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	Administrator logs in to their Code-Ko account with appropriate privileges.		
2	Administrator navigates to the user management or account administration section of the Code-Ko platform.		
3	Administrator selects the user for whom permissions need to be granted.		
4	Administrator views the available permissions.		
5	Administrator selects the appropriate permissions or permission set to assign to the user.		
6	Administrator confirms the selection and initiates the permission assignment process.	7	The system updates the user's account with the assigned permissions.
		8	The system displays a confirmation message indicating that the permissions have been successfully assigned to the user.

9	The user can now access the designated features or perform the specified actions based on their assigned permissions.		
Alternative Flow			
	If there are any issues during the permission assignment process, such as validation errors or conflicts.		The system displays an error message and advises the administrator to review and correct the selections.

3.4.24 Manage chatbot:

Name		Manage chatbot	
Actors		Admin	
Summary		This use case describes the process of managing the chatbot on the Code-Ko platform. The chatbot can be configured, updated, and customized by the administrator.	
Pre-Conditions		The administrator must have the necessary privileges and permissions to manage the chatbot. The administrator must be logged in to their Code-Ko account.	
Post-Conditions		The chatbot on the Code-Ko platform is successfully managed, including its configuration.	
Special Requirements		The platform should have a built-in chatbot feature or integration that can be managed by the administrator.	
Basic Flow			
Actor Action		System Response	
1	Administrator logs in to their Code-Ko account with appropriate privileges.		
2	Administrator navigates to the chatbot management section of the Code-Ko platform.		
3	Administrator views the current configuration and settings of the chatbot.		
4	Administrator can customize the chatbot's behavior, responses, or conversational flow based on the platform's available options.		
5	Administrator can update the chatbot's knowledge base, adding or removing questions, answers, or predefined responses.		
6	Administrator can train the chatbot using machine learning techniques to		

	improve its performance and accuracy.		
7	Administrator saves the changes or updates made to the chatbot's configuration and settings.		
Alternative Flow			
	If there are any issues during the chatbot management process, such as configuration errors or technical limitations.		The system displays an error message and advises the administrator to review and correct the settings.

3.4.25 Approve Content:

Name	Approve Content		
Actors	Admin		
Summary	This use case describes the process of reviewing and approving content uploaded by teachers on the Code-Ko platform. The administrator has the authority to ensure that the content meets the platform's guidelines and standards before it is made available to students.		
Pre-Conditions	The administrator must have the necessary privileges and permissions to review and approve teacher content. The administrator must be logged in to their Code-Ko account.		
Post-Conditions	The teacher content is successfully reviewed and approved by the administrator. Approved content is made available for students to access on the Code-Ko platform.		
Special Requirements	The platform should have a content management system that allows administrators to review and approve teacher content.		
Basic Flow			
Actor Action		System Response	
1	Administrator logs in to their Code-Ko account with appropriate privileges.		
2	Administrator navigates to the content management section of the Code-Ko platform.		
3	Administrator views the list of uploaded content by teachers that require review and approval.		
4	Administrator selects a piece of content for review.		
5	Administrator compares the content against the platform's standards and criteria for approval.		

6	Administrator makes a decision based on the evaluation: a. If the content meets the criteria, administrator approves the content. b. If the content does not meet the criteria, administrator provides feedback to the teacher for revision or rejects the content.		
7	If the content is approved: a. Administrator updates the status of the content to "Approved" in the content management system.		
8	Administrator proceeds to review and approve the next piece of content or completes the content review process.		
Alternative Flow			
	If the content is rejected: a. Administrator provides feedback to the teacher explaining the reasons for rejection.		

3.4.26 Reject Content:

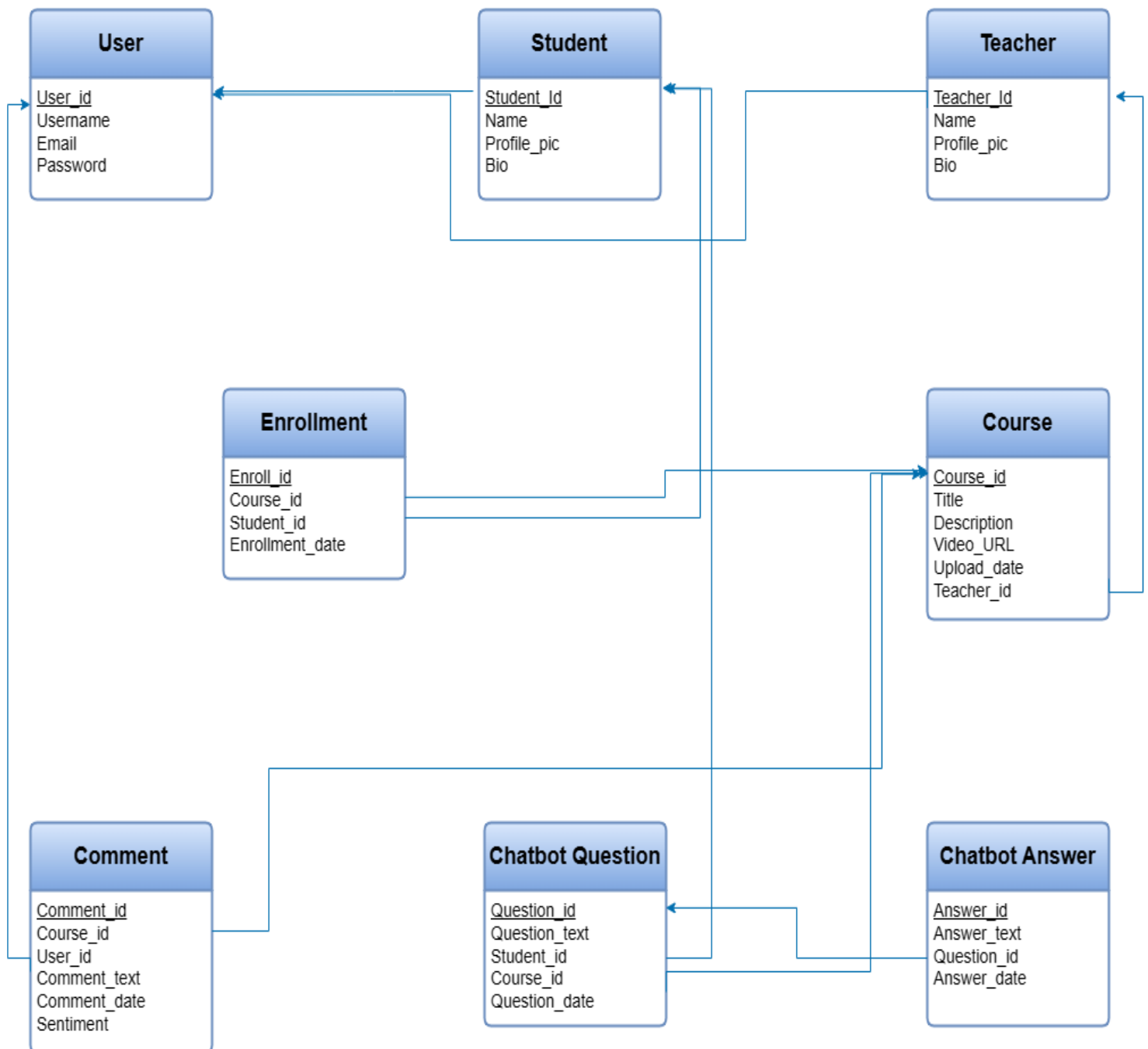
Name		Reject content	
Actors		Admin	
Summary		This use case describes the process of rejecting content uploaded by teachers on the Code-Ko platform.	
Pre-Conditions		The administrator must have the necessary privileges and permissions to reject content. The administrator must be logged in to their Code-Ko account.	
Post-Conditions		The rejected content is marked as rejected and the teacher is notified of the rejection. The teacher may revise the content and resubmit it for review.	
Special Requirements		The platform should have a content management system that allows administrators to review and reject teacher content.	
Basic Flow			
Actor Action		System Response	
1	Administrator logs in to their Code-Ko account with appropriate privileges.		
2	Administrator navigates to the content management section of the Code-Ko platform.		

3	Administrator views the list of uploaded content by teachers that require review.		
4	Administrator selects a piece of content for review.		
5	Administrator compares the content against the platform's standards and criteria for approval.		
6	If the content does not meet the criteria for approval: Administrator provides feedback to the teacher explaining the reasons for rejection.		
7	Administrator marks the content as "Rejected" in the content management system.	8	The rejected content is not made available to students on the Code-Ko platform.
9	Administrator proceeds to review the next piece of content or completes the content review process.		

3.4.27 Review Bad Reports:

Name		Review Bad Reports	
Actors		Admin	
Summary		This use case describes the process of reviewing and addressing bad reports submitted by users on the Code-Ko platform. The administrator is responsible for assessing the reports and taking appropriate actions to resolve the reported issues.	
Pre-Conditions		• The administrator must have the necessary privileges and permissions to review and address bad reports. • The administrator must be logged in to their Code-Ko account.	
Post-Conditions		The reported issues are reviewed and addressed by the administrator.	
Special Requirements		The platform should have a reporting system that allows users to submit reports regarding inappropriate content.	
Basic Flow			
Actor Action		System Response	
1	Administrator logs in to their Code-Ko account with appropriate privileges.		
2	Administrator navigates to the reporting management section of the Code-Ko platform.		
3	Administrator views the list of bad reports submitted by users.		
4	Administrator selects a report for review.		
5	Administrator accesses the details of the reported issue, including the reported content, user information, and any additional comments.		
6	Administrator evaluates the reported issue.		
7	If the report is valid and involves a violation of platform rules Administrator takes appropriate actions, such as removing the reported content, warning or suspending the user.		
8	Administrator proceeds to review the next bad report or completes the review process.		
Alternative Flow			
	If the report is invalid or does not involve a violation of platform rules Administrator dismisses the report.		

3.5 Database Schema diagram:



Chapter 4: Implementation and Test Cases

For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.

4.1 Implementation

Whatever implementation that you have done so far, please elaborate here.

Give clear details of the algorithms that were implemented along with the platform and the APIs which were used. **For FYP-1, this chapter can be changed to description of prototype developed.**

4.1.1 Implementation of First Component/Algorithm

Write implementation of first component of your system here.

4.2 Test case Design and description

This section will be added in FYP-II. Summarize the common attributes of test cases. This may include input constraints that must be true for every input in the set of associated test cases, any shared environmental needs, any shared special procedural requirements, and any shared case dependencies. The following scheme is recommended for describing test cases in detail.

4.2.1 Sample Test case No.1

<Software component Name>			
<Reference>			
Test Case ID:	<i>Reference Number</i>	Test Date:	<i>Date</i>
Test case Version:	<i>Version number</i>	Use Case Reference(s):	<i>Relation to use cases</i>
Revision History:	<i>Refer to previous test case identity (if any)</i>		
Objective	<i>Need and scope of the testing</i>		
Product/Ver/Module:	<i>Refer to overall system being built and the place of this test case in it.</i>		
Environment:	<i>Necessary and desired properties of the test environment. (hardware/software)</i>		
Assumptions:	<i>Assumptions that might affect the testing process.</i>		
Pre-Requisite:	<i>Necessary condition that needs to be fulfilled prior to the test case.</i>		
Step No.	Execution description	Procedure result	
	<i>Events being tested.</i>	<i>Mention software response.</i>	
Comments:			
☐ <i>Passed</i> ☐ <i>Failed</i> ☐ <i>Not Executed</i>			

4.2.2 Sample Test case No.2

- .
- .
- .

4.3 Test Metrics

Summarize here the common ground of attributes of test case metrics.

4.3.1 Sample Test case Metric.No.1

Metric:	Purpose
Number of Test Cases:	Total number of test cases that you have developed for your system.
Number of Test Cases Passed:	The number of test cases that successfully passed
Number of Test Cases Failed:	The number of test cases that failed
Test Case Defect Density:	(No of test cases failed * 100) No of test cases executed
Test Case Effectiveness:	No of defects detected using test cases *100 Total number of defects detected
Traceability Matrix:	Traceability is the ability to determine that each feature has a source in requirements and each requirement has a corresponding implemented feature.

4.3.2 Sample Test case Metric.No.2

4.3.3 Sample Test case Metric.No.3

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Chapter 5: Experimental Results and Analysis

This chapter will be added in FYP-II. Give proper analysis and discussion of experimental results (in plain English text) along with tables of results. **For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.**

Chapter 6: Conclusion and Future Directions

This chapter is mandatory. Give conclusions and summary of the work done. What were your findings and what were the results? Discuss in detail whether the scope of your project was entirely covered or not and whether the objectives of the project were met or not. What challenges did you face and what has been left out and why?

Sum up all the conclusions of all the chapters here to make a conclusion chapter. Do not repeat any text, just summarize it in different words.

Give recommendations for future work also. How your project can be further enhanced or improved? Future recommendations if someone wants to work on it. **For FYP-1 it is mandatory to list down a plan of the work to be done for FYP-2.**

References

List all important sources of information which have been consulted for this project

Appendix

Appendix A: Guidelines

This section should include all supporting information from the project that was not included in the body of the report. You should include surveys, complex statistical calculations, certain detailed tables and other such information in an appendix. The information presented in this section is important to support the work presented in the body of the report but would make it more difficult to read and understand if presented within the body of the report.

Cite the appendix items in the report narrative (write "see Appendix A") and organize appendices (e.g., Appendix A, Appendix B,

Any tables, figures, forms, or other materials that are not totally central to the analysis but that need to be included are placed in the Appendix.

Appendix B: Heading of Sample Appendix B

Following is a sample code with “code” style format.

```
Void Sample Function () {  
    Print "Hello World.";  
}
```

Formatting Guidelines

This document also serves as style guide for final year project reports. In order to give a similar high-quality appearance to all final year software project reports this template uses a collection of predefined Microsoft Word formatting styles. **These styles should be used without modification or replacement.** Font in the document is *“Time New Roman”*. This template provides following styles:

- **Title** – the main title style
- **Title2** – the subtitle style
- **Body Text** – style for paragraphs
- **Caption** – the style for a figure or table caption
- **Table Description** – the style for description of table, it must be added after caption.
- **Figure Description** - the style for description of figure, it must be added after caption.
- **Code** – the style for program source code

```
int x = 10; // Writing important code
```
- **Table Header Row** – Style for the header row of table
- **Table Grid** – the style for the data rows in the tables
- **Reference** – The style for references
- **Bullets** – The style for the bullet lists
- **Numbered List**– Style for numbered lists

All Heading styles with different level numbers are listed below.

Chapter 1: Heading 1

1.1 Heading 2

1.1.1 Heading 3

1.1.1.1 Heading 4

1.1.1.1.1 Heading 5

1.1.1.1.1 Heading 6

1.1.1.1.1.1 Heading 7

1.1.1.1.1.1 Heading 8

1.1.1.1.1.1.1 Heading 9

Tables and Figures

Tables and figures should be centered horizontally. The caption button should be used to insert caption for both the figures and tables. All figures and tables must be numbered properly. Always refer to tables and figures according to their numbers. A table or figure can be cited as follows: ‘see Table1’ or ‘as shown in Table1’. The caption of table should be centered above the table and figure caption should be centered below the figure. Place the tables/figures close to their reference. Use “Table Header Row” and ‘Table Grid’ style for table’s header and data rows respectively. It is compulsory to provide brief description of table/figure after its caption. Styles for table and figure descriptions are “Table Description” and “Figure Description” respectively.

Press Ctrl+Shift+S to see list of styles mentioned above. Figure 1 shows the Apply Style window displaying the list of styles. Select any text then press Ctrl+Shift+S, the Apply Style window will show you the current style applied on that text and if required, you can change the style by selecting any other style from the “Style Name” dropdown.

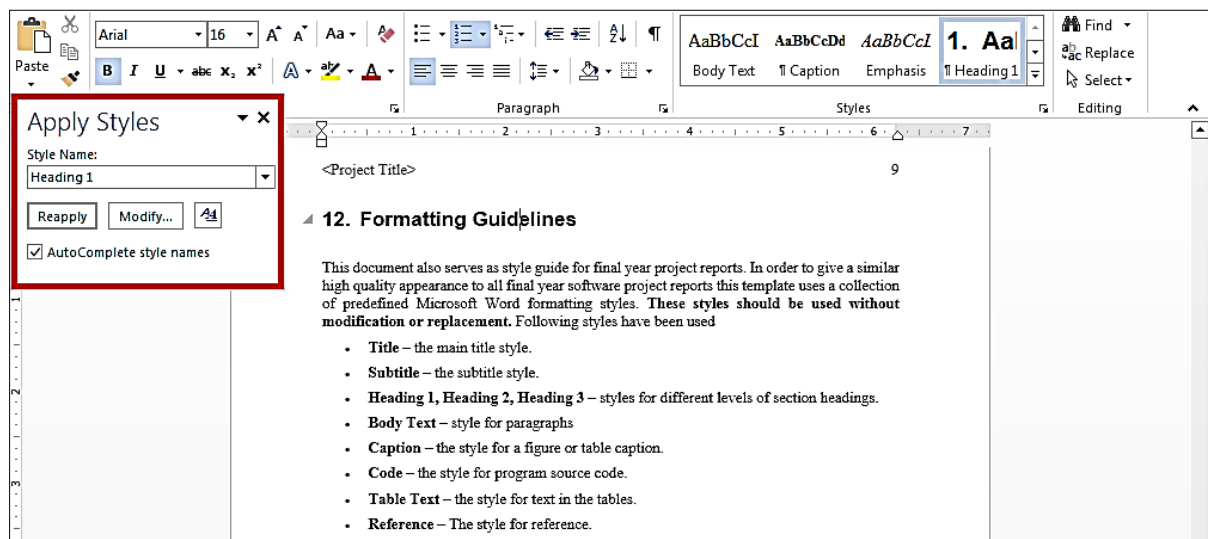


Figure 1: List of Styles
This is brief description of above figure.

Table 2: This is Sample table caption
This is brief description of following Table.

Header row	Header row	Header row	Header row
Row1 col1	Row1 col2	Row1 col3	Row1 col4
Row2 col1	Row2 col2	Row2 col3	Row2 col4

Table 3: This is Sample table caption
This is brief description of following Table.

Header row	Header row	Header row	Header row
Row1 col1	Row1 col2	Row1 col3	Row1 col4
Row2 col1	Row2 col2	Row2 col3	Row2 col4

Equations

Use equation editor to write equations in this report. Use last button of the custom tool bar to invoke equation editor. Similar to tables and figures, equations should also be aligned centered horizontally. Number all equations and insert them in parenthesis. Below is a sample equation and its reference number. An equation can be referenced like this: ‘it is clear from (1)’.

$$\sum_{\forall v \in V(G)} \deg(v) = 2|E(G)| \quad (1)$$

Header/Footer

Notice the headers in this document, before Introduction (i.e., the main content of this document) page numbers are in roman numerals. The page numbers of the actual content start with Arabic numerals i.e., 1, 2, 3 and so on. All of the **odd numbered pages** contain title of your project while the **even numbered pages** contain the section heading (i.e., chapter’s name) in the headers.

Other Formatting Guidelines

- Keep 2-4 GUIs in one page. Consume as much space as possible. Do not leave most of page blank unnecessarily.
- Do not break tables (or use cases) in multiple pages unless the table is too large to fit in one page.
- Re-arrange the content i.e., text, images, and tables properly to meet above two guidelines.

References

Always refer to the source of information by inserting the reference number in square brackets like this [5]. The reference numbers can either be added at the end of the sentence or within the sentence without changing the punctuation of sentence. A reference can also be cited as follows: ‘as Ruskey [2] mentioned’. List each source only once on your reference page.

- | |
|--|
| <p>[1] B. Klaus and P. Horn, <i>Robot Vision</i>. Cambridge, MA: MIT Press, 1986.</p> <p>[2] L. Stein, “Random patterns,” in <i>Computers and You</i>, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.</p> <p>[3] R. L. Myer, “Parametric oscillators and nonlinear materials,” in <i>Nonlinear Optics</i>, vol. 4, P. G. Harper and B. S. Wherret, Eds. San Francisco, CA: Academic, 1977, pp. 47-160.</p> <p>[4] M. Abramowitz and I. A. Stegun, Eds., <i>Handbook of Mathematical Functions</i> (Applied Mathematics Series 55). Washington, DC: NBS, 1964, pp. 32-33.</p> <p>[5] E. F. Moore, “Gedanken-experiments on sequential machines,” in <i>Automata Studies</i> (Ann. of Mathematical Studies, no. 1), C. E. Shannon and J. McCarthy, Eds. Princeton, NJ: Princeton Univ. Press, 1965, pp. 129-153.</p> <p>[6] Westinghouse Electric Corporation (Staff of Technology and Science, Aerospace Div.), <i>Integrated Electronic Systems</i>. Englewood Cliffs, NJ: Prentice-Hall, 1970.</p> <p>[7] M. Gorkii, “Optimal design,” <i>Dokl. Akad. Nauk SSSR</i>, vol. 12, pp. 111-122, 1961 (Transl.: in L. Pontryagin, Ed., <i>The Mathematical Theory of Optimal Processes</i>. New York: Interscience, 1962, ch. 2, sec. 3, pp. 127-135).</p> <p>[8] G. O. Young, “Synthetic structure of industrial plastics,” in <i>Plastics</i>, vol. 3, <i>Polymers of Hexadromicon</i>, J. Peters, Ed., 2nd ed. New York: McGraw-Hill, 1964, pp. 15-64.</p> |
|--|

Figure 2: IEEE Reference style

This figure represents the styling information for adding references in IEEE format

Following is a list of sample reference for various typed of sources in IEEE format.

- [1] P.M. Morse and H. Feshback, *Methods of Theoretical Physics*. New York: McGraw Hill, 1953. **//Format for Book**
- [2] S.K. Kenue and J.F. Greenleaf, "Limited angle multifrequency diffraction tomography," *IEEE Trans. Sonics Ultrason.*, vol. SU-29, no. 6, pp. 213-217, July 1982. **//Format for Journal Article**
- [3] B. Tsikos, "Segmentation of 3-D scenes using multi-modal interaction between machine vision and programmable mechanical scene manipulation," Ph.D. dissertation, Univ. of Pennsylvania, BCE Dept., Philadelphia, 1987. [Add if applicable: University Microfilms, Inc., University of Michigan, Ann Arbor, Michigan.] **//Format for Dissertation or thesis**
- [4] R. Finkel, R. Taylor, R. Bolles, R. Paul, and J. Feldman, "An overview of AL, programming system for automation," in *Proc. Fourth Int. Joint Conf Artif. Intell.*, pp. 758-765, Sept. 3-7, 1975. **//Format for Proceedings paper**
- [5] "Technology threatens to shatter the world of college textbooks, *The Wall Street Journal*, vol 91, pp. A1, A8, June 1, 1993. **//Format for Newspaper article**
- [6] R. Cox and J. S. Turner, "Project Zeus: design of a broadband network and its application on a university campus," Washington Univ., Dept. of Comp. Sci., Technical Report WUCS-91-45, July 30, 1991. **//Format for Technical Report**
- [7] M. Janzen, *Instant Access Accounting*. Computer software. Nexus Software, Inc IBM-PC, 1993. **//Format for Software**
- [8] Fuminao Okumura and Hajime Takagi, "Maglev Guideway on the Yamanashi Test Line," <http://www.rtri.or.jp/rd/maglev2/okumura.html>, October 24, 1998. **//Format for World Wide Web (give author and title if named)**
- [9] "AT&T Supplies First CDMA Cellular System in Indonesia," <http://www.att.com/press/1095/951011.nsa.html>, Feb 5, 1996. **//Format for World Wide Web**